Low-Window Prime-Tail Compression and Radical Certificates

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Abstract

For the actual centered theta–Gamma–prime operator we give a quantitative finite-rank approximation of its complete prime tail on any bounded spectral window. An explicit Gamma-frequency cutoff makes all parameters in the error budget available once the stated input bounds are supplied. Continuous filters and finite, true weighted-Gram calculations yield safe upper certificates for both tail signs after any finite number of known-radical constraints. The same construction gives an unconditional upper bound on the total multiplicity of the complete operator's spectrum below a fixed negative threshold. No low-window sign, numerical certificate, full-kernel identification, or proof or disproof of the Riemann hypothesis is claimed. The rank bound is very large, and compactness alone gives no rate adequate for a relative estimate at zero.

Keywords: theta kernel; prime-tail compression; finite-rank approximation; spectral windows; weighted Gram matrices.

1 The actual form and inputs

For $x \geq 0$ define $\tau_j(x) = \pi j^2 e^{2x}$ and the even theta kernel

$$\kappa(x) = e^{x/2} \sum_{j \geq 1} (4\tau_j^2 - 6\tau_j) e^{-\tau_j}, \quad W = \kappa/(2\cosh(x/2)), \quad \mathcal{H} = L^2_{\text{even}}(dx/W).$$

All Hilbert spaces are complex; inner products are conjugate-linear in the first argument. Let $m(u) = \int \cosh(x/2) u(x) dx$, $\mathcal{H}_c = \ker m$, and $Cu = u - 2m(u)\kappa$, the $\mathcal{H}$-orthogonal projection onto $\mathcal{H}_c$. Write $\|\cdot\|_2$ for ordinary Lebesgue norm and use the unitary ordinary Fourier transform. The complete centered form is

$$q_0[u] = t_\Gamma[u] - \langle u, Bu \rangle_{\mathcal{H}}, \quad t_\Gamma[u] = \int_{\mathbb{R}} d_\Gamma(t) |\widehat{u}(t)|^2 dt, \quad d_\Gamma(t) = 2 \int_0^\infty \frac{e^{-s/2}(1 - \cos ts)}{1 - e^{-2s}} ds,$$
$$d_\Gamma(t) \geq \tfrac{1}{2} \log(1 + 4t^2), \qquad d_\Gamma(t) = \log|t| + O(1) \quad (|t| \to \infty).$$

We use the following inputs for this precise actual operator from the accompanying working manuscript [1]: $X = \{u \in \mathcal{H} : t_\Gamma[u] < \infty\}$ is a closed form domain, B is bounded self-adjoint on $\mathcal{H}_c$ with $\|B\| \leq d$ for a known finite cap d, and L_c is the self-adjoint representative of q_0 on $X_c = X \cap \mathcal{H}_c$. The source supplies the explicit choice $d = 10$. The complete correction B contains $c_\Gamma \|u\|_2^2$ and *every* prime-power shift; $c_\Gamma = \log \pi - \psi(1/4)$. The lower Gamma bound follows directly from $1 - e^{-2s} \leq 2s$; the logarithmic asymptotic is also a standard digamma consequence [5]. We have $L_c \geq -dI$.

The same source establishes that $r_j = \kappa^{(2j)} - 4^{-j}\kappa$ are linearly independent full-domain radicals. Thus $\mathcal{N} = \overline{\operatorname{span}\{r_j\}}^{\mathcal{H}} \subseteq \ker L_c$, and $L_c P_{\mathcal{N}} = 0$ by closedness. Set $\mathcal{K} = \mathcal{H}_c \ominus \mathcal{N}$ and let $P_0 = \mathbf{1}_{\{0\}}(L_c)$. We do not assume $P_{\mathcal{N}} = P_0$.

For integer $M \geq 1$, the actual tail form is

$$p_{>M}[u] = 2\,\mathrm{Re} \sum_{n>M} \frac{\Lambda(n)}{\sqrt{n}} \int u(x + \log n)\overline{u(x)}dx.$$

It is represented by a bounded self-adjoint operator D_M on $\mathcal{H}_c$, obtained by centering its raw weighted row. The complete theta envelope and weighted shifts give [3]

$$W(x) \leq 40e^{4|x|-\pi e^{2|x|}}, \qquad W_* := \|W\|_\infty \leq 40e^{-\pi}, \qquad \|D_M\| \leq \tau_M \leq 160(M+1)^2 e^{-\pi(M+1)}, \quad (1)$$

where $\tau_M = 80\sum_{n>M} \Lambda(n) n^{3/2} e^{-\pi n}$. Wherever W_* enters an upper bound or parameter choice, the explicit cap $40e^{-\pi}$ may replace it. These are bounds on the actual infinite tail, not a claim that the tail is positive. We quote the proof inputs and source locations in Section 6.

2 Finite-rank approximation on a bounded window

Let $J \subset [-R, R]$ be Borel and bounded, $R \geq 0$, and $P_J = \mathbf{1}_J(L_c)$. Every $u \in \operatorname{Ran} P_J$ belongs to $\operatorname{Dom} L_c$, and

$$t_\Gamma[u] \leq (R+d)\|u\|_{\mathcal{H}}^2, \qquad \|u\|_2 \leq \sqrt{W_*}\|u\|_{\mathcal{H}}. \quad (2)$$

For integers $N \geq M$ let $D^{\mathrm{raw}}_{(M,N]}$ denote the finite sum of the two raw shifts $a_n W(x) u(x \pm \log n)$, $a_n = \Lambda(n)/\sqrt{n}$. Each individual oriented row is bounded from ordinary L^2 into the *ambient* weighted space, because $\|W(\cdot)u(\cdot + s)\|_{\mathcal{H}} \leq \sqrt{W_*}\|u\|_2$. Pair the shifts before asserting even output, and then apply C on $\mathcal{H}$. Consequently

$$\|C D^{\mathrm{raw}}_{(M,N]}\|_{L^2_{\mathrm{even}} \to \mathcal{H}_c} \leq C_{M,N} := 2\sqrt{W_*} \sum_{M<n\leq N} \frac{\Lambda(n)}{\sqrt{n}}. \quad (3)$$

The tail $n > N$ is bounded on $\mathcal{H}$ by τ_N. We do not claim that the full infinite tail is bounded from ordinary L^2 to $\mathcal{H}$.

Let F_T be the ordinary Fourier projection to $[-T, T]$, $W_B = \sup_{|x|>B} W(x)$, and $\nu(T) = \inf_{|t|>T} d_\Gamma(t)$. For $B, T > 0$ and $m \geq 1$, define the even finite-rank map

$$\mathcal{T}_{B,T,m} u = \mathbf{1}_{[-B,B]}(x) \sum_{\substack{0\leq k<m \\ k \text{ even}}} x^k \ell_k(u), \qquad \ell_k(u) = \frac{1}{\sqrt{2\pi}\, k!} \int_{-T}^{T} (it)^k \widehat{u}(t) dt. \quad (4)$$

Only ordinary L^2 membership of the output is used in the approximation argument. Its rank is at most $\lceil m/2 \rceil$. Set

$$\zeta_{B,T,m} = \sqrt{\frac{2BT}{\pi}} \frac{(BT)^m}{m!}, \qquad b_{R;B,T,m} = \sqrt{W_B} + \sqrt{\frac{R+d}{\nu(T)}} + \sqrt{W_*}\,\zeta_{B,T,m}. \quad (5)$$

Theorem 2.1. *For $\nu(T) > 0$, let $A = C D^{\mathrm{raw}}_{(M,N]} \mathcal{T}_{B,T,m}|_{\mathcal{H}_c}$. Then* $\operatorname{rank} A \leq \lceil m/2 \rceil$ *and*

$$\|(D_M - A)P_J\| \leq E := \tau_N + C_{M,N} b_{R;B,T,m}. \quad (6)$$

With $S = (A + A^*)/2$, the controlled self-adjoint compression is

$$\|P_J(D_M - S)P_J\| \leq E. \quad (7)$$

The corresponding unsymmetrized row bound for $(D_M - S)P_J$ is not asserted.

Proof. By (2), ordinary spatial tails outside $[-B,B]$ have norm at most $\sqrt{W_B}\|u\|_{\mathcal{H}}$, and Fourier tails outside $[-T,T]$ at most $\sqrt{(R+d)/\nu(T)}\|u\|_{\mathcal{H}}$. For real tx the order-m Taylor remainder of e^{itx} is bounded by $|tx|^m/m!$. The Hilbert–Schmidt norm on $[-B,B]\times[-T,T]$ is at most $\zeta_{B,T,m}$. On even inputs the odd terms vanish, so

$$\|u-\mathcal{T}_{B,T,m}u\|_2 \le b_{R;B,T,m}\|u\|_{\mathcal{H}} \quad (u\in \operatorname{Ran} P_J).$$

The finite rows are bounded by (3) and the discarded full prime tail by τ_N, proving (6). On $\operatorname{Ran} P_J$, symmetrization replaces $P_J(D_M-A)P_J$ by its self-adjoint part, which cannot increase its norm. This proves (7). □

The columns of A are centered finite prime-shift rows of the ordinary compact functions $\mathbf{1}_{[-B,B]}x^k$; a sharp spectral projection is not used to form A. For fixed J, suitable choices below make $E\to 0$, so D_MP_J is compact even when P_J has infinite rank.

3 An explicit cutoff and rank schedule

Fix $0<\varepsilon\le 1$. Choose

$$N=\max\{M,\lceil 3+\log(4/\varepsilon)\rceil\}, \tag{8}$$

so $\tau_N<\varepsilon/4$ by (1). If $C_{M,N}=0$, the choice $A=0$ already suffices. Otherwise fix any explicit cap $C_\sharp\ge C_{M,N}>0$, put

$$X_\varepsilon=\max\{e^2,4C_\sharp\sqrt{40}/\varepsilon\},\quad Y_\varepsilon=\frac{4}{\pi}\log X_\varepsilon,\quad B=\tfrac12\log Y_\varepsilon, \tag{9}$$

and use the following *explicit* Gamma cutoff, including the case $R+d=0$:

$$H_\varepsilon=\max\left\{1,\frac{16C_\sharp^2(R+d)}{\varepsilon^2}\right\},\qquad T=\frac12\sqrt{e^{2H_\varepsilon}-1}. \tag{10}$$

Since $d_\Gamma(t)\ge\frac12\log(1+4t^2)$, $\nu(T)\ge H_\varepsilon>0$ and the frequency term in (6) costs at most $\varepsilon/4$. Finally choose m to be the smallest positive integer satisfying both inequalities

$$m\ge 4eBT,\qquad m\ge\max\left\{0,\log_4\left(\frac{4C_\sharp\sqrt{W_*}\sqrt{2BT/\pi}}{\varepsilon}\right)\right\}. \tag{11}$$

Then $E\le\varepsilon$. Indeed $Y_\varepsilon\le X_\varepsilon$ and the decreasing envelope give $C_\sharp\sqrt{W_B}\le C_\sharp\sqrt{40}Y_\varepsilon e^{-\pi Y_\varepsilon/2}\le\varepsilon/4$. Also $m!\ge(m/e)^m$ gives $\zeta\le\sqrt{2BT/\pi}\,4^{-m}$ and the last term is at most $\varepsilon/4$.

For a no-prime-search cap, the elementary Chebyshev bound used in [2] gives

$$C_{M,N}\le 2\sqrt{W_*}\,[4(\log 2)\sqrt{N}+5]. \tag{12}$$

For the Chebyshev function $\psi_{\mathrm{C}}(x)=\sum_{n\le x}\Lambda(n)$, this follows from the binomial estimate $\psi_{\mathrm{C}}(n)\le\psi_{\mathrm{C}}(\lceil n/2\rceil)+n\log 2$, iterated to $\psi_{\mathrm{C}}(x)\le 2(\log 2)x+\log x+\log 2$, and partial summation. Choose $C_\sharp$ to be the right-hand side of (12), not an arbitrarily inflated cap, when estimating the rank. For fixed M,R,d,W_*, (8)–(11) then give

$$\operatorname{rank} A\le\lceil m/2\rceil\le\exp\Big(O\big(\varepsilon^{-2}[1+\log(1/\varepsilon)]\big)\Big). \tag{13}$$

The constants in $O(\cdot)$ depend on the fixed data. This upper bound uses the specified smallest admissible m; it need not hold for an arbitrarily larger choice. It is an upper bound from one explicit construction, not an optimality or practicality claim. The factor e^{2H_ε} in (10) is essential: it is the inversion of the displayed logarithmic Gamma lower bound.

4 Filtered finite Gram certificates

Let $I \subset J$, and let a real continuous $0 \le \phi \le 1$ equal one on I and be supported in J. Set $F = \phi(L_c)$ and $S_\phi = \frac{1}{2}F(A + A^*)F$. Since $FP_J = F$, (6) implies the *global* filtered estimate

$$\|FD_MF - S_\phi\| \le E. \tag{14}$$

Write the finite-rank operator as $A = YG^*$, where G consists of Riesz representatives $g_k \in \mathcal{H}_c$ for $\ell_k|_{\mathcal{H}_c}$ and Y of the corresponding output columns. They are actual weighted-space columns; for even k one may take $g_k = C(Wb_k)$ with $\widehat{b_k}(t) = (-it)^k \mathbf{1}_{[-T,T]}(t)/(\sqrt{2\pi}k!)$. Put

$$Z = (FY, FG), \qquad H_0 = \frac{1}{2}\begin{pmatrix} 0 & I \\ I & 0 \end{pmatrix}, \qquad S_\phi = ZH_0Z^*.$$

Let P_j be the true $\mathcal{H}$ projection onto $\operatorname{span}\{r_1, \dots, r_j\}$, $Q_j = I - P_j$, and $R_j = (r_1, \dots, r_j)$. With the positive definite true radical Gram $H_j = R_j^*R_j$ and $B_j = R_j^*Z$, the filtered Gram is

$$G_j = Z^*Q_jZ = Z^*Z - B_j^*H_j^{-1}B_j \ge 0. \tag{15}$$

The nonzero spectrum of $Q_jS_\phi Q_j$ equals that of $G_j^{1/2}H_0G_j^{1/2}$ away from zero; a singular G_j is allowed. Therefore, for every $u \in \mathcal{K}$ with $Fu = u$,

$$-\big(E + \max\{0, -\lambda_{\min}(G_j^{1/2}H_0G_j^{1/2})\}\big)\|u\|_{\mathcal{H}}^2 \le p_{>M}[u] \le \big(E + \max\{0, \lambda_{\max}(G_j^{1/2}H_0G_j^{1/2})\}\big)\|u\|_{\mathcal{H}}^2. \tag{16}$$

For an empty represented matrix the extrema are taken as zero. These are safe *absolute* bounds for either sign, not a certified sign. Since $P_j \to P_{\mathcal{N}}$ strongly and S_ϕ has finite rank, $Q_jS_\phi Q_j \to P_{\mathcal{K}}S_\phi P_{\mathcal{K}}$ in norm. The nonnegative Rayleigh extrema in (16) decrease with j, but no effective rate is claimed. Actual column and Gram errors must be bounded and added to E; no such numerical calculation is claimed here.

The sharp projection P_J is used only in the proof of the error estimate. The filter F remains the complete operator's functional calculus, not a Gamma-only substitute. If desired, one can approximate it without a sharp-projection oracle: for $s > d$, let $R_s = (L_c + s)^{-1}$ and $h(r) = \phi(1/r - s)$ for $r > 0$, $h(0) = 0$. A degree-q Bernstein polynomial of h on $[0, (s-d)^{-1}]$ yields

$$\|p_q(R_s) - F\| \le \frac{\operatorname{Lip}(\phi)(s+R)^2}{2(s-d)\sqrt{q}} \tag{17}$$

when ϕ is Lipschitz and supported in $[-R, R]$. This follows from $\operatorname{Var}(K/q) \le 1/(4q)$ for a binomial K. The polynomial has zero constant term. Actual resolvent actions still require certification: for $v \in \operatorname{Dom} L_c$,

$$\|v - R_sf\|_{\mathcal{H}} \le \frac{\|(L_c + s)v - f\|_{\mathcal{H}}}{s - d}.$$

If $\widetilde{F}$ satisfies $\|\widetilde{F} - F\| \le \eta$, replacing F in S_ϕ costs at most $\eta(2 + \eta)\|A\|$. These bounds give a certificate format, not computed columns.

5 Kernel distinction and negative spectral count

For a fixed bounded J containing a neighborhood of zero, let $E_\delta = \mathbf{1}_{[-\delta,\delta]\setminus\{0\}}(L_c)$. Then $E_\delta \to 0$ strongly, and compactness of D_MP_J gives $\|D_ME_\delta\| \to 0$. This is only an absolute $o(1)$ statement, not

$O(\delta)$ or $o(\delta)$. For continuous filters ϕ_δ equal to one on $[-\delta,\delta]$, supported in $[-2\delta,2\delta]$, and bounded by one, the known-radical compression instead has the norm limit

$$P_{\mathcal{K}}\phi_\delta(L_c)D_M\phi_\delta(L_c)P_{\mathcal{K}} \longrightarrow (P_0 - P_{\mathcal{N}})D_M(P_0 - P_{\mathcal{N}}). \tag{18}$$

The limit may be nonzero when the entire kernel contains directions beyond $\mathcal{N}$. Both limits follow by strong convergence of the projections or filters and compactness of the bounded-window tail rows. Neither supplies a relative low-collar tail sign.

Theorem 5.1 (Negative spectral count)**.** *For $0 < \delta \le \min\{1, d\}$, the total multiplicity of the spectrum of L_c in $[-d,-\delta]$ is at most $\lceil m/2\rceil$, where the explicit construction* (9)–(11) *is used with $\varepsilon = \delta/2$, $R = d$, and the* complete-correction *cap and cutoff specified in the proof. In particular,*

$$\dim \operatorname{Ran} \mathbf{1}_{[-d,-\delta]}(L_c) \le \exp\Big(O_{d,W_*,c_\Gamma}(\delta^{-2}[1+\log(1/\delta)])\Big).$$

For $\delta > d$ the spectral interval is empty. No assertion is made that the count is positive or zero.

Proof. In the preceding construction replace the tail by the *entire* correction B, choose $N = \max\{1, \lceil 3+\log(4/\varepsilon)\rceil\}$, and take the finite raw numerator $c_\Gamma W u$ plus both orientations of all prime powers $2 \le n \le N$, followed by C. Its ordinary-to-weighted cap is

$$C_N^{\mathrm{full}} = \sqrt{W_*}\left(|c_\Gamma| + 2\sum_{2\le n\le N}\frac{\Lambda(n)}{\sqrt{n}}\right) \le \sqrt{W_*}\,[|c_\Gamma| + 8(\log 2)\sqrt{N} + 10].$$

The omitted part has $\mathcal{H}$ norm at most τ_N. Set $C_\sharp$ to this displayed upper bound and use (9)–(11), with $R = d$. This produces a finite-rank A_{full} satisfying $\|(B - A_{\mathrm{full}})P_{-\delta}\| \le \varepsilon = \delta/2$, where $P_{-\delta} = \mathbf{1}_{[-d,-\delta]}(L_c)$.

If $\dim \operatorname{Ran} P_{-\delta} > \operatorname{rank} A_{\mathrm{full}}$, some nonzero u in that spectral range satisfies $A_{\mathrm{full}}u = 0$. The spectral theorem gives $q_0[u] \le -\delta\|u\|_{\mathcal{H}}^2$, while $t_\Gamma \ge 0$ and the row estimate give $q_0[u] \ge -\varepsilon\|u\|_{\mathcal{H}}^2$, a contradiction. The stated rank cost follows from (13) and the complete-correction cap. Every negative spectral vector is automatically orthogonal to the entire kernel, but no kernel projection is needed in this argument. □

The theorem excludes negative essential spectrum and permits negative eigenvalue accumulation only at zero [6]. It does not decide the sign of L_c or supply coordinates for its negative spectral subspace. At a relative target $\varepsilon = c\delta$, the rank schedule is extremely expensive, and neither (16) nor compactness proves that its finite matrix contribution is $O(\delta)$.

6 Source and verification boundary

The supplied working manuscript [1] establishes the theta and operator normalization, the complete bounded correction, the Fourier lower bound, and the actual derivative-radical identities (notably Lemmas a:lem:theta, a:lem:prime, a:lem:fourier, and a:lem:radical). The accompanying research notes [2, 3, 4] record the immediate construction and tail estimates; they are not independent published references. The code package checks elementary algebra and parameter schedules, but does not certify any spectral filter, resolvent column, Gram entry, numerical sign, or complete-operator inequality. We make no novelty or priority claim. The low-collar sign and the Riemann hypothesis remain unresolved by these results.

References

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